

PROVIDENCE HARBOR, RI, USA

WMO: 997278

Lat: 41.807N

Lon: 71.401W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.8	N/A	N/A	N/A	N/A	N/A	N/A	11.4	0.3	10.5	-1.6	5.0	330	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	32.2	N/A	30.4	N/A	28.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	9.3	8.2	-14.5	34.9	2.8	1.1	-16.5	35.7	-18.2	36.4	-19.7	37.0	-21.8	37.8		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	-1.1	0.2	3.9	10.1	15.3	20.3	24.4	23.1	19.2	13.4	7.4	3.3
	DBStd	9.49	5.36	4.63	4.48	3.68	3.99	3.69	2.61	2.76	3.40	3.92	4.37	4.94
	HDD10.0	1187	343	277	194	42	3	0	0	0	0	13	101	214
	HDD18.3	2922	602	509	447	247	113	21	0	1	30	159	327	466
	CDD10.0	1803	1	1	6	47	166	309	445	406	275	118	23	7
	CDD18.3	496	0	0	0	1	18	80	187	149	55	6	0	0
	CDH23.3	3251	0	0	1	12	122	504	1362	964	264	22	0	0
	CDH26.7	936	0	0	0	5	31	163	427	252	57	2	0	0
Wind	WSAvg	3.8	3.8	4.0	4.5	4.5	4.2	3.9	3.7	3.4	3.2	3.6	3.5	3.7
Precipitation	PrecAvg	1154	100	88	109	102	91	83	74	94	98	96	108	110
	PrecMax	1715	296	183	224	324	213	281	205	256	201	391	280	273
	PrecMin	646	13	10	14	38	18	4	10	18	23	10	10	26
	PrecStd	193	55	37	52	52	45	57	41	55	49	70	53	58
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	15.3	19.7	24.1	30.2	34.0	34.1	33.7	30.9	25.7	19.9	18.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	12.6	11.0	15.1	20.2	26.4	30.6	32.4	31.0	28.2	23.2	17.7	15.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	9.4	8.5	12.5	17.9	24.0	28.3	30.7	29.3	26.2	21.1	15.8	13.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	6.7	6.5	10.3	16.0	21.8	26.1	29.1	27.9	24.6	19.6	14.3	11.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	7.5	7.6	7.8	8.3	8.4	8.0	7.9	7.7	8.0	7.7	7.4	6.9
		MCDBR	11.1	10.3	10.8	11.2	12.0	11.3	9.9	9.5	9.7	9.1	9.1	9.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.297	0.306	0.331	0.372	0.413	0.459	0.482	0.440	0.374	0.338	0.311	0.297	
	taud	2.496	2.479	2.463	2.379	2.297	2.205	2.150	2.254	2.417	2.530	2.564	2.559	
	Ebn at Noon	859	906	919	897	864	822	797	821	860	859	839	832	
	Edh at Noon	73	87	99	116	129	142	149	130	103	82	68	64	
All-Sky Solar Radiation	RadAvg	1.86	2.73	3.73	4.80	5.30	5.79	6.00	5.39	4.38	2.94	2.01	1.55	
	RadStd	0.13	0.25	0.27	0.51	0.54	0.57	0.33	0.28	0.32	0.27	0.15	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	+45

Nomenclature: See separate page